

# Boxed: A Docker-Based Code-Execution Substrate for Autonomous Code-Generating Agents

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**Abstract**—Autonomous large language model (LLM) agents write, run, and revise code inside their reasoning loop, so the sandbox is on the hot path of every task. Roll-your-own Docker is quick but is usually deployed without a read-only root, capability drops, process caps, or egress control. Hosted sandbox services are hardened but move the operator’s code and data to a third party. Research virtual machine monitors isolate well but are built for a boot-once serverless model. Boxed is an open-source substrate for agentic code execution: a self-hosted Go control plane, a 1.32 MiB Rust in-sandbox agent that streams stdout, stderr, and files over JSON-RPC 2.0, and a driver interface whose four lifecycle methods admit backends beyond the Docker driver that exists today. Measured against the Docker Engine API on the same host, image, and command, the control plane and agent cost 16% of a raw Docker lifecycle, on a laptop and on an idle server, and the hardened configuration is faster than stock Docker defaults because dropping the network costs less than setting up the bridge. With the OCI runtime swapped under the unchanged driver, a lifecycle takes 354ms with runc, 405ms with gVisor, and 7824ms with a Kata microVM, and the substrate’s own cost stays close to constant. A twelve-vector escape probe scored by post-conditions read from the host denies all twelve under runc and gVisor and eleven under Kata, whose guest kernel lacks the host policy that had been stopping a ptrace attach on the in-sandbox agent; a one-line change to the agent brings Kata to twelve. Giving the microVM an internal, inter-container-isolated network and no CPU quota cuts its lifecycle to 2856ms with all twelve still denied. On the same host, the OpenHands agent-server sandbox in its SDK’s default configuration takes 7.6s per lifecycle and stops 9 of the twelve vectors; the cloud metadata endpoint, and with it the instance’s service-account token, is reachable from the workload. Throughput on one host is bound by the Docker daemon: four times the cores gives  $1.2\times$  the sandboxes per second. Boxed, the harness, and every raw trace are MIT-licensed.

**Index Terms**—sandboxing, code execution isolation, large language models, autonomous agents, containers, Docker, JSON-RPC

## I. Introduction

Autonomous code-generating agents arrived in force between 2024 and 2026, from Yang et al.’s SWE-agent [1] and the AutoGPT-style harnesses to products such as Cursor, Devin, and Claude Code, and they changed what “running untrusted code” means in practice. Executable code has been argued to be a better action space for an agent than text or JSON [2]. An agent built that way emits dozens of short programs per task: it writes a snippet, runs it, reads stdout, reads back a file it produced, and tries

again. The sandbox sits inside that loop, and its latency is paid on every iteration.

An operator who wants that loop hosted has three options today, and each gives something up.

Roll-your-own Docker is the common choice: quick to set up and routinely shipped without seccomp profiles, egress filters, a read-only root filesystem, or process and memory caps, which leaves the host one rm -rf / away from compromise [3]; measurement studies report the same pattern at scale [4], and cgroup-based resource isolation can itself be evaded [5]. Hosted sandbox services (E2B [6], Modal Sandbox [7], Cloudflare Sandbox [8]) ship hardened isolation, and in exchange every byte of generated code, output, and artifact leaves the operator’s infrastructure for a third party, which some deployment rules do not allow. Research VMMs such as Firecracker [9] have the right primitives but are built for AWS Lambda’s boot-once model and have no in-guest agent to stream artifacts back to the caller.

What agent workloads need is a substrate: an isolation backend chosen at deployment time, an agent inside the sandbox that streams standard output and on-disk artifacts as they appear, and a deployment the operator runs, with the control plane and the workload data under their control.

## A. Problem Statement

An autonomous coding agent runs untrusted, machine-generated code many times in a single task, and three things have to hold at once. The isolation boundary has to contain code nobody reviewed. Per-execution latency has to stay low enough that the sandbox does not dominate the agent’s loop. And the operator has to keep control of the keys and the data, because agent workloads carry proprietary source and customer records into the sandbox as a matter of course.

No existing substrate satisfies all three. Hosted services meet the first two, but they put a third party inside the trust boundary and give the operator no say over the isolation backend. Self-hosted container runtimes leave the operator in charge but tie the control plane to one isolation technology, so a stronger boundary later means rewriting the orchestration layer. Boxed targets that gap: keep the

control plane, the keys, and the data with the operator, and make the isolation backend a deployment-time choice.

Two questions follow, and Section VI answers them by measurement. What do the control plane and agent cost over calling Docker directly on the same host, image, and command? And once the Docker driver applies the hardening that roll-your-own deployments skip, which of a standard set of container-escape vectors does it stop, judged by the state left after each attempt and not by strings in the attempt’s output?

## B. Objectives

- An isolation interface narrow enough that container runtimes, microVMs, and Wasm sandboxes can be interchanged behind it without changing the control plane.
- Incremental delivery of output and of files produced during execution, as events on one channel.
- Self-hosted deployment with no Boxed-operated service in the data path.
- Measured cost and limits against a raw-Docker baseline, failures included.

## C. Contributions

- A Go driver interface whose four lifecycle methods (Create, Start, Stop, Connect) are the contract a backend has to satisfy, so that the isolation backend becomes a deployment-time setting. The release implements one backend, Docker. No microVM or Wasm backend exists yet, and this paper claims only what the Docker driver does.
- A Rust in-sandbox agent that streams stdout and stderr chunks as discrete JSON-RPC events instead of buffering until process exit, mirrors files written under a watched directory back to the caller as artifact events, and reports the child’s real exit status.
- A self-hosted control plane authenticated by an operator-configured API key. It authenticates one instance and does not manage credentials or tenants.
- A released benchmark harness and campaign: lifecycle latency against raw Docker through the Engine API on the same host; throughput under concurrency with repeated sweeps and confidence intervals; idle per-sandbox overhead; a twelve-vector escape probe scored by post-condition checks read from the host; and an end-to-end agent trace on official HumanEval tasks with the model, prompts, token counts, and exit codes recorded. Raw CSV and JSONL traces ship with the source.
- A comparison of post-condition scoring with the signature matching that escape probes commonly use, on the same attempts: signature matching misreports as failures exactly the vectors where enforcement kills or silences the process.
- The same driver, agent, and harness under three OCI runtimes on one host (runc, gVisor, Kata),

which measures the backend switch the interface was designed for, exposes and closes a ptrace gap that the container runtime had been masking, and shows that two hardening settings free under runc cost seconds under a microVM, together with the driver setting that removes that cost.

- The OpenHands agent-server sandbox measured with the same harness on the same host, and a core-scaling sweep on 4, 8, and 16 vCPUs: single-host throughput is bound by the daemon.

## II. Background and Related Work

### A. Container-based isolation

Linux containers, built on namespaces, cgroups, and seccomp, underpin nearly every development sandbox. They start fast when images are warm (90 ms for a stock docker run on our host, Table II), so they dominate workloads that create and destroy environments often; the cost is a shared host kernel. CVE-2019-5736 [3] is the standard reminder that one bug in runc turns a container into the host. Lin et al. [4] ran 88 of 223 collected exploits inside a default-configured container: 50 succeeded, capabilities, seccomp, and MAC stopped more privilege escalations than namespaces and cgroups did, and 11 exploits still broke isolation. That ordering is the reason the hardening in Section V-B starts with the capability drop. Hardened runtimes such as gVisor [10] put a userspace kernel between the workload and the host; Young et al. [11] measure its system-call cost at  $2.8\times$  native in its fastest mode,  $9\times$  when the call reaches the host, and  $72\times$  through the Gofer, with file opens  $216\times$  slower. Anjali et al. [12] compare Linux containers, gVisor, and Firecracker on CPU, network, memory, and file system microbenchmarks and trace the kernel code each executes; both run substantially more kernel code than native Linux, and neither wins on every workload. Docker can select gVisor or Kata as a runtime, which Section VI-H measures.

nsjail, bubblewrap, and firejail apply the same namespace, seccomp, and cgroup primitives to a single process, with no image and no daemon. For an operator who does not need Docker’s image ecosystem they are the natural comparison. Boxed uses Docker for two reasons: agent workloads are specified as images (a Python runtime, a toolchain, a set of packages), and the Engine API provides the exec-attach stream the in-sandbox agent runs over. A process-sandbox backend behind the same interface is possible but is not evaluated here.

### B. Lightweight virtual machines

Agache et al. [9] showed that a stripped-down KVM VMM can boot a guest kernel in  $\sim 125$  ms with  $< 5$  MiB of memory overhead per VM, and Firecracker now powers AWS Lambda. Kata Containers [13], which Section VI-H measures as a Docker runtime, and Manco et al. [14] pursue the same hypervisor-as-isolation thesis. These systems

optimize boot cost and steady-state memory for a boot-once serverless pattern; a coding agent creates, uses, and destroys a sandbox many times per task, so the whole lifecycle is the cost that matters.

### C. Unikernels and WebAssembly

Madhavapeddy et al. [15] compile the application together with a library operating system into a single boot image, minimizing the trusted computing base at the cost of application-specific compilation, which rules out arbitrary dynamically generated binaries. WebAssembly [16] and WASI [17] make the guest portable bytecode with capability-based imports and the sandbox a runtime in the host process; microsecond instantiation is what a create-heavy agent workload wants, but agents emit unmodified Python scripts, shell commands, and native binaries that assume a POSIX environment, which the WASI toolchain does not yet run unmodified.

### D. Agent-oriented sandbox platforms

Several products now target LLM agents specifically. E2B [6] runs Firecracker microVMs behind a SaaS API and reports about 150ms cold start; Modal Sandbox [7] layers gVisor over Firecracker; Daytona [18] ships an AGPL, Docker-backed, self-hosted development environment; Cloudflare Sandbox [8] reuses V8 isolates for about 5ms cold start and limits execution to JavaScript and Wasm. OpenHands [19] executes every agent action in a Docker container through a REST action-execution API inside it, the closest design to Boxed’s driver-plus-agent split, but it is an agent framework with a sandbox inside and has no backend interface. The vendors’ cold-start figures were measured under undisclosed conditions; Table I repeats them as reported. Across the category the platform fixes the isolation mechanism, and apart from Daytona and OpenHands the platforms are hosted services.

### E. Evaluating LLM-generated code execution

Chen et al. [20] made single-function code generation a standard benchmark, and Yang et al. [1] extended it to agents that have to run the code they write. Our agent trace (Section VI-F) follows that regime on official HumanEval tasks and splits wall-clock into model inference and sandbox lifecycle, which is how much of an agent’s latency a substrate can affect at all.

## III. Threat Model

The threat model assumes an adversary with full control over the code submitted to a sandbox: they may craft arbitrary Linux binaries, issue any syscall sequence, and attempt network egress to any destination. The adversary does not control the control plane, the driver implementation, or the host kernel; these remain trusted.

The design aims to prevent four classes of attack.

T1, host filesystem access. Read or write access to the host filesystem outside the sandbox’s ephemeral root,

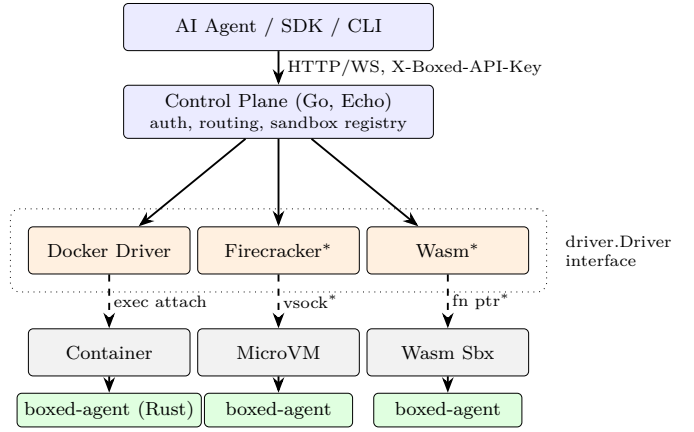


Fig. 1. Boxed architecture. The control plane drives one Driver interface. The Docker driver is the only implementation; the Firecracker and Wasm boxes show where a microVM or Wasm backend would attach. A common Rust agent runs inside every sandbox and speaks JSON-RPC over whatever byte stream the driver supplies. \*Not implemented; shown as a design target only.

whether through path manipulation, symbolic links, or mount operations.

T2, namespace escalation. Escape from the allocated network or process namespace into the host’s network stack or process tree, or tampering with the in-sandbox agent from a workload process that shares its namespace.

T3, co-tenant exfiltration. Access to credentials, artifacts, or data belonging to a concurrent sandbox on the same host.

T4, resource exhaustion. Denial of service against the host through fork bombs, memory exhaustion, or other unbounded resource consumption.

Side channels through shared hardware are out of scope; defending against them means per-customer hardware partitioning, an operator decision. The escape probe in Section VI-E covers T1, T2, and T4 directly and T3 only through the network vectors, since the sandbox has no route to a co-tenant except the network; no vector tries a cross-sandbox read through the host.

## IV. Design

Boxed has three components and one narrow interface (Fig. 1): a stateless control plane, a driver layer, and an in-sandbox agent, with orchestration kept apart from the isolation mechanism so that one control plane can drive Docker on a laptop and a microVM in production. Today only the Docker driver exists.

### A. Driver Interface

The central abstraction is one Go interface. In the current source it has twelve methods; eight are helpers for file injection, listing, health, and naming that the SDK and control plane use. The four lifecycle methods below are the minimum a backend has to implement, and the control plane’s sandbox path depends on nothing else:

TABLE I

Agent-oriented sandbox runtimes (data as of 2026-05). Cold-start figures for systems other than Boxed are vendor claims measured under undisclosed conditions and are reproduced as reported, not validated by us; the Boxed figure is the end-to-end create+exec+destroy median of Section VI-B and includes an exec, which the vendor figures may not.

| System                 | Isolation            | Cold start        | Backend selectable | Self-host | License     | First-class artifacts |
|------------------------|----------------------|-------------------|--------------------|-----------|-------------|-----------------------|
| Boxed (this work)      | Docker (runc)        | 178 ms (measured) | interface only*    | yes       | MIT         | yes                   |
| E2B [6]                | Firecracker          | ~150 ms (claim)   | no                 | limited   | Apache-2.0  | yes                   |
| Modal Sandbox [7]      | gVisor + Firecracker | ~1.5 s (claim)    | no                 | no        | proprietary | yes                   |
| Daytona [18]           | Docker               | not reported      | no                 | yes       | AGPL        | no                    |
| Cloudflare Sandbox [8] | V8 isolates          | ~5 ms (claim)     | no                 | no        | proprietary | limited               |

\*The driver interface is designed for other backends; Docker is the only implementation in this release.

```

type Driver interface {
  Create(ctx context.Context,
    cfg SandboxConfig) (string, error)
  Start(ctx context.Context, id string) error
  Stop(ctx context.Context, id string) error
  Connect(ctx context.Context,
    id string) (io.ReadWriteCloser, error)
  // ... file, listing, and health helpers
}

```

Create builds a sandbox from the configuration and returns an identifier; Start and Stop bound its lifetime; Connect returns the raw byte stream the control plane uses to talk to the in-sandbox agent (a Docker exec attach today; a vsoc connection or a host-side function for other backends). The control plane does not know which it has. Whether four methods are enough for a microVM backend is untested, since none has been written, and is listed as future work.

## B. Control Plane

The control plane is a stateless HTTP and Web-Socket server on the Echo framework that maps POST /v1/sandbox requests through the configured driver. create returns as soon as the runtime reports the sandbox process launched (for Docker, when ContainerStart acknowledges), with status: ready; the client never polls, and the first exec absorbs the wait for the agent’s JSON-RPC loop. An agent task may create dozens of sandboxes, and one round trip saved per sandbox adds up. Authentication is a static, operator-configured API-key header (X-Boxed-API-Key); Boxed does not issue, rotate, or store keys, and this authenticates one instance without multi-tenant identity, key management, or compliance. Generated code, execution output, and intermediate files stay on infrastructure the operator controls.

## C. In-Sandbox Agent

boxed-agent runs inside every sandbox: one Rust binary on tokio, serde, and notify (1.32 MiB stripped; it links the C library dynamically and needs nothing else at runtime) speaking JSON-RPC 2.0 as newline-delimited JSON over whatever byte stream the driver supplies. On exec the agent spawns the command as a child process and streams stdout and stderr chunks back as separate RPC events as

they arrive, so a model can react to partial output or an early error before the program finishes. When the child exits the agent reports its exit status. At startup the agent marks itself non-dumpable (PR\_SET\_DUMPABLE), for the reason Section VI-H gives. Alongside, an inotify watcher monitors the artifact directory (/output). Any file written there is emitted as an artifact event with the file’s MIME type and base64 contents, so an agent that asks Python to write plot.png gets the bytes pushed to it, with no second round trip and no polling.

## D. Artifact Protocol

Files are first-class outputs because much of what a model asks code to do shows up as a side effect on disk and never reaches the console. The current release streams them inline: the watcher reads the file, guesses its MIME type, base64-encodes it, and emits it on the same channel as stdout and stderr. A 10 MiB cap bounds the memory and serialization cost; larger files are skipped with a logged warning. The wire protocol reserves a url field on the artifact event for an out-of-band path, in which the agent would PUT the file to a pre-signed URL and return a reference, and that path is not implemented. A substrate that silently drops large artifacts is a correctness hazard for any workload that emits checkpoints or datasets. Both the protocol and the agent sit above the driver interface, so artifact semantics do not depend on the backend.

## V. Implementation

The release is 2,485 lines of Go in cmd and internal (control plane, Docker driver, CLI) and 960 lines of Rust for the agent; the binaries are 12.0 MiB and 1.32 MiB stripped.

## A. Agent injection and container lifecycle

The Docker driver bind-mounts the host’s boxed-agent binary into the container read-only at create time, so a stock base image (python:3.10-slim) can be reused across sandboxes with no Boxed-specific image registry to maintain. The image root filesystem is read-only. Separate tmpfs mounts back /tmp, /output, and the configured working directory, so user-written data is ephemeral and never touches the host filesystem.

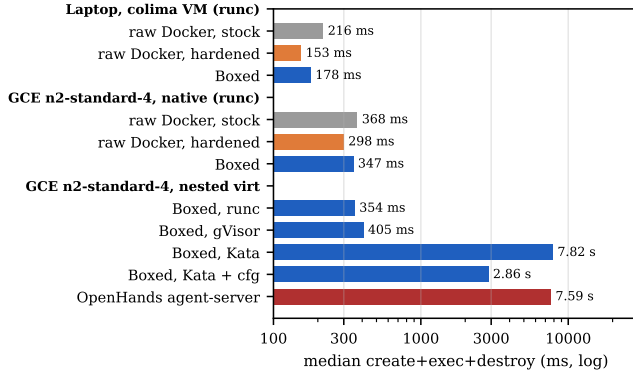


Fig. 2. Median create→exec→destroy for every configuration measured, grouped by host. Same image and command throughout; Sections VI-B, VI-G, VI-H, and VI-I give the sample sizes and confidence intervals.

The container’s own PID 1 is a `tail -f /dev/null` placeholder whose only job is to keep the namespace alive. Connect launches boxed-agent as a Docker exec, and the attached stream carries JSON-RPC. This matters for security: the agent is a child of the container’s init, so any workload process in the sandbox’s PID namespace can see it, and the ptrace vector in Section VI-E targets the agent for that reason. Under a microVM backend the agent would be the guest init and talk over vsock.

## B. Resource and network model

Every sandbox is created with a CPU quota (NanoCPUs), a memory cgroup, and a PidsLimit of 256. The container root is read-only, all Linux capabilities are dropped, and no-new-privileges is set. Docker’s default seccomp profile applies; Boxed does not yet ship its own. The resource defaults are 1.0 core and 512 MiB, with ceilings of 4.0 cores and 8 GiB. A TTL goroutine stops the sandbox when its timeout expires.

The Docker backend always uses Docker’s none network, so only loopback exists. Requests that enable networking or name an allow-list are rejected until Boxed has an enforceable per-sandbox egress policy. This fails closed. It does not claim that Docker’s default bridge would be an adequate egress control.

## VI. Evaluation

This section answers the two questions from Section I: what the control plane and agent cost over calling Docker directly, and which escape vectors the hardened driver stops under post-condition scoring. It also reports throughput under concurrency, idle per-sandbox overhead, and an end-to-end agent trace. Every number below expands from a macro that the released analysis scripts generate from the raw traces in the repository.

Fig. 2 collects every lifecycle measured in the paper on one axis.

TABLE II

Lifecycle latency on two hosts, medians over 5 runs × 200 sequential create→exec→destroy cycles per configuration, same image and command. Raw Docker rows call the Engine API directly with no control plane and no agent. All values in ms.

| Configuration                                | create | exec | destroy | total | IQR | p95 |
|----------------------------------------------|--------|------|---------|-------|-----|-----|
| MacBook Pro M1 Pro, colima VM, laptop in use |        |      |         |       |     |     |
| Raw Docker, stock                            | 90     | 53   | 73      | 216   | 44  | 499 |
| Raw Docker, hardened                         | 76     | 46   | 29      | 153   | 42  | 379 |
| Boxed (plane + agent)                        | 98     | 48   | 30      | 178   | 33  | 327 |
| GCE n2-standard-4, native Linux, idle        |        |      |         |       |     |     |
| Raw Docker, stock                            | 172    | 94   | 102     | 368   | 13  | 386 |
| Raw Docker, hardened                         | 153    | 80   | 65      | 298   | 9   | 310 |
| Boxed (plane + agent)                        | 197    | 83   | 66      | 347   | 10  | 360 |

## A. Setup

The laptop measurements were taken on a MacBook Pro (Apple M1 Pro, 16 GB, macOS 26.5) with Docker Engine 29.5 (linux/arm64, kernel 6.8, cgroup v2, seccomp and AppArmor on) inside a 4-vCPU, 8 GiB colima virtual machine on Apple’s Virtualization framework. The python:3.10-slim image was pre-pulled. The machine ran its usual background processes (editor, browser) throughout and was not quiesced; we report dispersion across repeated runs so that the contention shows. Unless noted otherwise each lifecycle runs a synthetic `print('ok')` workload, which keeps application compute out of the substrate measurement. The Boxed control plane was built from the tagged release with Go 1.24 and the agent with Rust 1.80.

## B. Lifecycle latency against raw Docker

The substrate’s cost only means something next to what Docker costs on the same host, so three configurations ran the same create→exec→destroy sequence. Raw Docker, stock defaults calls the Engine API with docker run’s default host configuration. Raw Docker, hardened calls the Engine API with exactly the host configuration Boxed’s driver applies (read-only root, CapDrop ALL, no-new-privileges, PID limit, network none, 1 CPU, 512 MiB, tmpfs mounts) and has no control plane or agent. Boxed goes through the HTTP API. The raw configurations run `python3 -c "print('ok')"` through a Docker exec; Boxed runs the same command through the in-sandbox agent. Each configuration ran 5 runs of 200 sequential lifecycles, interleaved so that drift on the host hits all three alike.

Table II (upper half) gives the result. Boxed’s median end-to-end lifecycle is 178 ms (bootstrap 95% CI 177–179 ms; per-run medians 172 ms to 190 ms). The interquartile range is 33 ms, and the tail reaches 327 ms at p95 and 578 ms at p99. Raw Docker with the same hardened configuration takes 153 ms. Its tail is heavier than Boxed’s (p95 379 ms), but the tails are not a property of the configurations: of the 50 slowest raw-hardened lifecycles, 27 fall in a single run, and of the 50 slowest stock lifecycles, 25 fall in another, almost all of them in the create phase. Those are bursts of host activity that landed on whichever configuration was running at the time, so Boxed’s smaller

p95 is host timing. The difference of 25 ms (95% CI on the difference of medians 23–27ms; 16% of the raw cost) is what the HTTP control plane, the agent launch over exec attach, and the JSON-RPC exec path add together: 22 ms on create, 2 ms on the first exec, and 1 ms on destroy. It buys an HTTP API that any language can call, an agent that streams output and files as events, and a create that takes one round trip. An operator who already drives the Engine API from Go and does not need streaming artifacts would not pay it.

The two raw-Docker rows compare the hardening against stock defaults. Boxed’s hardened configuration is 63 ms faster end to end than docker run’s defaults. The stock configuration attaches the container to the default bridge network, and setting up and tearing down that attachment costs more than the tmpfs mounts and the capability drop do; with network none that work is gone. On this host the hardening that roll-your-own deployments skip does not cost latency.

### C. Throughput under concurrency

Client concurrency was swept over  $c \in \{1, 2, 4, 8, 16, 32\}$  with 80 create→destroy lifecycles per level, and the whole sweep was repeated 10 times. Throughput rises from 7.0 sandboxes/s at  $c=1$  to a plateau of 17.2 ( $\pm 2.7$ ) at  $c=16$  and 17.1 at  $c=32$ , with the last three levels inside one another’s confidence intervals. A single sweep at  $c=4$  or  $c=8$  can land anywhere in a two-to-one band, so a curve drawn from one sweep per level, as such curves usually are, is mostly noise. Section VI-J shows where the plateau comes from.

### D. Idle per-sandbox overhead

Each of 30 idle sandboxes was sampled five seconds after create returned. Median working set, as docker stats reports it from the container cgroup, is 0.43 MiB (p95 0.67 MiB); median CPU is 0.01%. This is cgroup accounting inside a VM, so we take it as an order of magnitude; it fits an agent that is a stripped Rust binary parked on one tokio loop.

### E. Escape probe with post-condition checks

The hardened driver was exercised with twelve attempts drawn from common container-escape patterns: a filesystem mount, reading the host through /proc/1/root, the Docker or containerd socket, loading a kernel module, a raw socket, a new user and PID namespace, egress to an RFC1918 address and to the link-local metadata endpoint, a fork storm, a memory bomb that touches every page, a write to the root filesystem, and PTRACE\_ATTACH on the in-sandbox agent. Each attempt runs in a fresh sandbox, and the whole suite ran 3 times.

Each vector gets two verdicts on the same attempt. The signature verdict is the method escape probes commonly use: capture stdout and stderr and grep for a denial string. The post-condition verdict looks at state after the

attempt: whether a mount appeared in /proc/mounts, whether the written file exists, the errno the syscall returned, which device and inode /proc/1/root resolves to, and, for the resource vectors, the cgroup’s pids.events and memory.events counters read on the host. Table III gives both verdicts and the evidence for every vector.

The post-condition checks classify 12 of 12 vectors as denied (100%), with 0 undenied and 0 harness errors, and the verdicts were identical across the 3 runs. The signature verdicts disagree on 2 of 12, and in each case they report as undenied an attempt the evidence shows was stopped.

The evidence column shows why. The cgroup kills the memory bomb and the host’s oom\_kill counter records it, but the process prints nothing before SIGKILL, so a signature reads that silence as a successful attack; the fork storm gets exactly the forks the PID limit allows and then EAGAIN; /proc/1/root resolves to the container’s own root; with network none the sandbox has a loopback interface and nothing else. The ptrace attach is denied because CapDrop ALL removes CAP\_SYS\_PTRACE and the host kernel’s Yama scope of 1 limits attachment to descendants; a host without Yama would need more, and Section VI-H finds exactly that host. The module-load vector calls init\_module directly because the image has no modprobe binary, and a probe that invoked one would measure the image. A probe scored by strings in captured output therefore understates coverage on precisely the vectors where enforcement kills or silences the process.

This is a behavioural probe over twelve vectors and does not prove isolation. It does not exercise kernel vulnerabilities, it runs one image, and it has no cross-sandbox read (T3). None of the twelve attempts escaped the host.

### F. End-to-end agent trace

A code-generation loop drove Boxed over the first 20 tasks of the official HumanEval release [20] in file order, 3 times, for 60 task executions. For each task the model (claude-opus-5, default adaptive thinking, 4096 max output tokens, Anthropic Messages API, no fallback model) gets the signature and docstring and returns the function; Boxed runs the prompt, the completion, and the task’s published check in a fresh sandbox, and the exit status decides. A deliberately wrong candidate exits 1 with an AssertionError, which is the harness’s control. Model, timestamp, prompt hash, token counts, stop reason, raw completion, program, and captured output for every task are in agent\_trace.jsonl next to each CSV.

60 of 60 task executions passed (100%; stop reasons end\_turn: 60). Median end-to-end wall-clock was 3.95s: the model call took 3.42s and the sandbox lifecycle 453 ms (per-repetition medians 436–504 ms). Summed over all executions, model inference is 85% of wall-clock (80–88% across repetitions) and the sandbox 15%. Sandbox create in this loop had a median of 279 ms (268–328 ms across repetitions), against 98 ms in the tight lifecycle loop of

TABLE III

Escape probe on the hardened Docker driver, 12 vectors, 3 independent runs (verdicts identical across runs). Signature is signature matching (grep for a denial string in captured output); Post-condition checks state after the attempt, on the host where possible. The paper reports the post-condition column.

| Vector       | Threat | Exit | Signature | Post-condition | Evidence                                                                                                  |
|--------------|--------|------|-----------|----------------|-----------------------------------------------------------------------------------------------------------|
| mount_host   | T1, T2 | 0    | denied    | denied         | mounts_on_/mnt=0                                                                                          |
| proc1_root   | T1     | 0    | denied    | denied         | proc1_root_dev_ino=42:1193209 root_dev_ino=42:1193209<br>pid1=tail<br>sockets_found=0 api_reachable=False |
| docker_sock  | T2, T3 | 0    | undenied  | denied         | init_module errno=1(Operation not permitted)                                                              |
| modprobe     | T2     | 0    | denied    | denied         | errno=1(Operation not permitted)                                                                          |
| raw_socket   | T2     | 0    | denied    | denied         | unshare rc=1 unshare: unshare failed: Operation not permitted                                             |
| setns        | T2     | 0    | denied    | denied         | ifaces=lo                                                                                                 |
| rfc1918      | T3     | 0    | denied    | denied         | ifaces=lo                                                                                                 |
| imds         | T3     | 0    | denied    | denied         | forked=250 EAGAIN after 250 forks host:pids.max=256<br>pids.events=max 1                                  |
| fork_bomb    | T4     | 0    | denied    | denied         | exit=-1 host:memory.max=536870912 oom_kill 1<br>/etc:absent /usr/bin:absent                               |
| mem_bomb     | T4     | -1   | undenied  | denied         | PTRACE_ATTACH pid=7 errno=1(Operation not permitted)                                                      |
| ro_root      | T1     | 0    | denied    | denied         | yama=1                                                                                                    |
| ptrace_agent | T2, T3 | 0    | denied    | denied         |                                                                                                           |

Section VI-B. During all three repetitions the host’s one-minute load average was between 13 and 17 on eight cores, from activity unrelated to the experiment; the lifecycle campaign ran under lighter load, and the recorded load is kept next to the traces. So the gap is an upper bound on what an idle interval between creates costs, and on a quieter host the model’s share would be larger. The pass rate measures the model; for the substrate, the trace shows how large a share of an agent iteration the sandbox takes with a current frontier model.

#### G. The same campaign on a native Linux host

To separate the cost of the virtual machine from the cost of the substrate, the whole campaign was repeated on an idle Google Compute Engine n2-standard-4 (4 vCPUs, 16 GB, Ubuntu 24.04, Docker Engine 29.7, kernel 6.17). Table II (lower half) gives the lifecycle result. Boxed’s median lifecycle is 347 ms against 298 ms for raw hardened Docker, a substrate cost of 48 ms (95% CI 48–49 ms; 16% of the raw cost), and the hardened configuration is 70 ms faster than stock. Throughput peaks at 12.3 sandboxes/s ( $\pm 0.1$ ) at  $c=8$  with 0 errors. The escape probe denies 12 of 12 vectors, identical across 3 runs, with the signature method disagreeing on 2. The agent trace passed 60 of 60 executions with model inference at 89% of wall-clock, and sandbox create in the agent loop had a median of 207 ms against 197 ms in the tight loop on the same host. On a quiet host the gap seen on the laptop disappears, as it should if load caused it.

The substrate’s share of the raw cost is 16% here and 16% on the laptop, the hardened configuration is faster than stock on both, and the escape verdicts match. Absolute latency does not carry over: this Xeon is slower per lifecycle than the M1 Pro under colima, and its dispersion is far smaller (IQR 10 ms against 33 ms), the difference between an idle server and a laptop in use.

#### H. One driver, three runtimes

Docker accepts alternative OCI runtimes per container, and the driver passes one through as a single field (BOXED\_DOCKER\_RUNTIME). Nothing else changes: the same control plane, agent, image, and harness ran under runc, under gVisor’s runsc (a user-space kernel), and under Kata Containers (a QEMU microVM with its own guest kernel) on one GCE n2-standard-4 with nested virtualization. This is the runtime-level version of the backend switch the interface was designed for. Table IV gives the result; Kata’s campaign was cut to  $2 \times 40$  lifecycles and one short sweep because each lifecycle boots a virtual machine.

Boxed’s lifecycle is 354 ms under runc, 405 ms under gVisor ( $1.1\times$ ), and 7824 ms under Kata ( $22.1\times$ ). The substrate cost is 50 ms, 64 ms, and 75 ms: close to a constant, so it is 14% of a container lifecycle, 16% of a gVisor one, and 1% of a microVM one. Most of a lifecycle is the boundary itself; the control plane and agent add a small, fixed amount on top. Kata’s raw stock lifecycle is 2.8 s on this host and its hardened one 7750 ms; isolating the flags (knobs.csv, three runs each), network none takes a bare Kata boot from 2.8 s to 7.7 s and the one-core CPU quota to 10.7 s, because the quota throttles QEMU while the guest boots. Two settings free under runc, where network none is faster than the bridge, cost seconds under a microVM. So the driver gained two settings: an operator-created Docker network that the driver verifies is internal with inter-container traffic disabled, used instead of none, and a switch that drops the CPU quota while keeping the memory and PID limits. The last two rows of Table IV measure them: under Kata the Boxed lifecycle falls from 7824 ms to 2856 ms ( $8.1\times$  runc) with 12 of 12 still denied, both egress vectors included. Throughput follows the same order (12.5, 9.5, 0.5 sandboxes/s, 1.2 with the settings); Kata also failed to create 23 of 72 sandboxes once concurrency reached 16, where the other runtimes had none.

TABLE IV

One driver, three OCI runtimes on one host (GCE n2-standard-4 with nested virtualization). Medians in ms over  $n=1000$  (runc),  $n=1000$  (runcsc),  $n=80$  (kata),  $n=200$  (kata-aware),  $n=100$  (runc-aware) sequential lifecycles; substrate cost is Boxed minus raw hardened Docker under the same runtime, with a bootstrap 95% CI. Peak is the best mean sandboxes/s over the sweeps.

Denied is the post-condition escape verdict, identical across 3 probe runs for every runtime;  $\rightarrow$  gives the count after the agent fix of Section VI-H, also 3 runs. + cfg: the microVM settings (internal ICC-off network, no CPU quota).

| Runtime    | stock | hardened | Boxed | cost (95% CI) | peak/s | denied                 |
|------------|-------|----------|-------|---------------|--------|------------------------|
| runc       | 376   | 304      | 354   | 50 (49–51)    | 12.5   | 12/12                  |
| gVisor     | 414   | 341      | 405   | 64 (63–65)    | 9.5    | 12/12                  |
| Kata/QEMU  | 2947  | 7750     | 7824  | 75 (63–89)    | 0.5    | 11/12 $\rightarrow$ 12 |
| Kata + cfg | 2960  | 2793     | 2856  | 63 (55–75)    | 1.2    | 12/12                  |
| runc + cfg | 311   | 292      | 328   | 36 (34–38)    | 10.2   | 12/12                  |

The escape probe is where the runtimes differ most. All twelve vectors are denied under runc and gVisor, and eleven under Kata, identical across three runs. Three of the verdicts depend on the probe reading the host instead of the sandbox, because a runtime with its own kernel changes what an attempt looks like from inside. Under gVisor and Kata, unshare succeeds, but the new namespace’s inode is absent from the host’s /proc: it lives inside the sandbox’s kernel, and the probe scores it by that host check. Under both, the fork storm and the memory bomb kill the whole sandbox, agent included, where under runc only the offending process dies. Kata’s guest kernel has no module loading at all (init\_module returns ENOSYS).

The one undenied vector is PTRACE\_ATTACH on the agent under Kata. Under runc that attach was denied by the host kernel’s Yama scope; Kata’s guest kernel has no Yama, the workload and the agent both run as root, and a same-uid attach needs no capability, so the attach succeeds. Under runc the host’s policy had been doing the work, and the stronger boundary took it away. The fix is one line: the agent now calls `prctl(PR_SET_DUMPABLE, 0)` at startup, after which the kernel’s own access check refuses a tracer without `CAP_SYS_PTRACE`, which the driver drops. Re-running the probe with that agent on the same host type gives 12 of 12 under Kata (3 runs, identical), with the attach failing `PTRACE_ATTACH pid=2 errno=1(Operation not permitted) yama=absent`, and 12 and 12 of 12 under runc and gVisor. The lifecycle numbers in this section predate that change; a `prctl` at startup is off the lifecycle path and was not re-measured.

#### I. A deployed competitor on the same host

OpenHands [19] is the closest design sibling in Section II and its agent-server sandbox is self-hostable, so the harness was pointed at it on the same host type: the container is started exactly as the OpenHands SDK’s DockerWorkspace starts it, the harness waits for its health endpoint, and commands go through its bash

TABLE V

OpenHands agent-server and Boxed (runc, hardened) on the same host type, same harness. Lifecycle medians in ms over  $n$  cycles; post-condition escape verdicts, identical across 3 runs.

| System                 | $n$  | create | first exec | destroy | total | denied |
|------------------------|------|--------|------------|---------|-------|--------|
| OpenHands agent-server | 200  | 6485   | 35         | 1068    | 7590  | 9/12   |
| Boxed, runc            | 1000 | 202    | 84         | 68      | 354   | 12/12  |

TABLE VI

Throughput against host core count (GCE n2-standard-{4,8,16}, native runc, create $\rightarrow$ destroy lifecycles, 80 per level). Mean sandboxes/s with 95% CI at the best level.

| vCPU | sweeps | at $c=1$ | peak           | at $c$ | at $c=32$ |
|------|--------|----------|----------------|--------|-----------|
| 4    | 10     | 3.8      | $12.3 \pm 0.1$ | 8      | 11.5      |
| 8    | 10     | 4.3      | $13.6 \pm 1.0$ | 8      | 11.2      |
| 16   | 10     | 4.4      | $15.2 \pm 0.1$ | 4      | 11.3      |

API (Table V). The lifecycle is 7.6s ( $21\times$  Boxed under runc), almost all of it the agent-server’s own startup, which includes a VS Code server; OpenHands creates one sandbox per session and runs many actions in it, so that cost is paid per session where Boxed pays per task. The probe results are the larger difference. In the SDK’s default configuration the sandbox runs as an unprivileged user, so the root-filesystem write fails, but it has no PID limit (forked=2000 no EAGAIN in 2000 forks host:pids.max=19108 pids.events=max 0), no memory limit, and a bridge network from which the cloud metadata endpoint answered: the probe read the instance’s hostname and a service-account access token from inside the sandbox. The unprivileged user also has passwordless sudo. These are the SDK’s defaults; OpenHands documents custom sandbox images for operators who need more.

#### J. Throughput against core count

The laptop plateau in Section VI-C could be the four cores or the one Docker daemon. To tell them apart, the throughput sweep was repeated ten times on native n2-standard-8 and n2-standard-16 hosts with the same image and harness (Table VI, Fig. 3). Peak throughput goes from 12.3 sandboxes/s on 4 vCPUs to 13.6 on 8 and 15.2 on 16:  $4\times$  the cores buys  $1.2\times$  the throughput, and on 16 vCPUs the peak arrives at  $c=4$  and falls back to about the 4-vCPU level by  $c=32$ . The single-client rate barely moves (3.8, 4.3, 4.4). Container churn on one host is bounded by the daemon’s serialization of create and destroy, and adding cores does not change that; a deployment that needs more sandboxes per second needs more daemons, which is the multi-host case in Section VIII.

#### K. Threats to validity

The laptop measurements ran inside a virtual machine on a machine in use; the native Linux campaign removes both conditions and reproduces the relative results, so

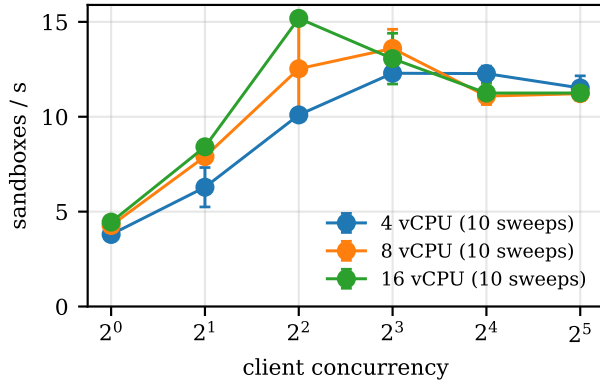


Fig. 3. Sandbox creation rate against client concurrency on 4, 8, and 16 vCPU hosts (native Linux, runc, 10 sweeps each, mean with 95% CI).

the comparison the paper rests on, Boxed against raw Docker with host, image, and command fixed, holds on both. Absolute latencies do not transfer between hosts. Each repeat ran the three configurations in a fixed order. The Kata numbers come from nested virtualization, which inflates a microVM boot; the runtime ordering and knob attribution survive that, the absolute Kata latency does not, and Kata’s  $n$  is 80 and 200. OpenHands was measured in its SDK’s default configuration; a deployment that follows its custom-sandbox guidance would score differently, and its per-session lifecycle is not the per-task one Boxed pays. The escape probe is twelve vectors on one image, and it tests configuration; it does not test kernel vulnerabilities. Throughput lifecycles are create→destroy without an exec, so they measure the daemon’s container churn rate and not the latency an agent sees. The figures for other systems in Table I are vendor claims we did not reproduce. The agent trace uses one model and twenty tasks, three times, on a loaded host.

## VII. Discussion

### A. What the runtime swap says about the design

Section VI-H is the closest this paper comes to testing the interface claim. The control plane and agent did not change and did not need to; a one-field change moved the substrate from a shared kernel to a user-space kernel to a guest kernel, and its own cost stayed flat. Two things did have to change. The hardening had been tuned against runc, and two of its settings were expensive under Kata until the driver gained the internal network and the quota switch (2.7× on the lifecycle, with the probe results unchanged). And the agent’s placement as a peer process, flagged in Section V, became a real ptrace exposure once the host kernel’s Yama policy was no longer underneath it: the stronger boundary exposed a gap that the host’s policy had been masking.

### B. Pool-based sandbox reuse

The driver creates a fresh container per request, which guarantees a clean start and pays the create phase every time; create is the largest phase in Table II and nearly all of a microVM lifecycle. Reusing a warmed sandbox for compatible follow-up requests would spread that cost, as Oakes et al.’s SOCK [21] does with an LRU cache of paused containers, and the interface already lets Create return an existing identifier; what reuse costs in isolation has to be measured first.

## VIII. Future Work

The limitations above set the order of work: applying the microVM settings automatically when the driver detects a microVM runtime, instead of by operator flag; a custom seccomp profile and an egress policy, so the network restriction is Boxed’s and not the host’s; sandbox reuse, measured for the latency it saves and the isolation it weakens; a Firecracker driver, which the runtime swap does not reach because Firecracker under Docker needs block-device rootfs storage; then a multi-host scheduler, since Section VI-J shows one daemon is the ceiling, and a Wasm/WASI backend once the toolchain runs unmodified language runtimes.

## IX. Conclusion

Boxed is a Docker-based execution substrate for LLM agents that create, use, and destroy sandboxes many times per task: a streaming in-sandbox agent, a one-round-trip create, and a self-hosted control plane behind an interface meant to admit stronger backends. The substrate costs 16% of a raw Docker lifecycle on a laptop and on an idle server alike, and the hardening it applies makes raw Docker faster than stock defaults on both. The same driver ran under runc, gVisor, and a Kata microVM with its cost flat across the three, the backend switch the interface was built for; the swap exposed a ptrace attach on the agent that only the host kernel’s policy had been stopping, since closed in the agent and re-verified under Kata, and two hardening settings that cost seconds under a microVM until the driver learned to replace them. Against the OpenHands sandbox on the same host, Boxed is 21× faster per lifecycle and denies three vectors that OpenHands’ defaults do not, one of them the cloud metadata endpoint. Scored by post-conditions read from the host, the hardened driver stops 12 of 12 escape vectors under runc; signature matching would have reported 2 of them as failures. Open items are in Sections VII and VIII. Implementation, harness, campaign runner, and all raw traces are MIT-licensed at <https://github.com/akshayaggarwal99/boxed> (tag v0.3.2-paper); make tables regenerates every number, table, and figure here.

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